

Designing for Solar

by Joseph McCabe

Deciding where to put a PV system on a roof can be a struggle, especially if the best place for the system is unavailable for some reason. That's when an architect or designer may need to consider integrating the PV system with the roof or with a porch overhang—and that's where BIPV Designer software can help. BIPV Designer lets users quickly see how much energy can be produced by putting building integrated photovoltaics (BIPV) on various building surfaces. By moving around in the graphical user interface, from simple to detailed, inexperienced users can learn the effects of placing BIPV in different areas. Seasoned professionals will also find the tool valuable. This simple yet powerful program is available for either Windows or Macintosh platforms and can be downloaded for free on the Internet. (Unfortunately, feedback indicates that it does not run on the most recent Mac OS.)

DESIGNING DETAILS

In 1996, Energy Ideas, my company, developed the first version of BIPV Designer on a floppy disk. We were then contracted by Solar Design Associates to expand this program to include more functions. Christopher Gronbeck of Sustainable by Design was instrumental in developing the new graphical user interface (GUI). Gronbeck also helped to write the Macromedia Director coding, especially the coding for the absorber surface shading algorithms.

When developing and refining the software, Energy Ideas concentrated on meeting the needs of designers and architects. The user interface was redesigned to have a look and feel that architects would appreciate. We modeled its design after the flamboyant style of *Metropolis* magazine, a publication for architects and design-



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ers. We also took care to structure the program's metric in terms that architects are familiar with. For example, architects care about area coverage, not necessarily electric-power production, so we express area in either square feet or square meters, rather than expressing it in terms of an energy measurement like kW or kWh.

In the spring of 2000, the shareware version of BIPV Designer became available on CD-ROM, and soon after that it became available on the Internet. We opened a Yahoo user group for discussions of the software, and provided technical support via e-mail. While technical support was not guaranteed for users, we made good-faith efforts to help if people had questions. We also developed a frequently asked questions (FAQ) page and a features and benefits evaluation page and made these available on the Internet.

BIPV Designer is divided into five sections. These deal respectively with 1) the overview, 2) the site location, 3) the building design specifics, 4) the PV system used, and 5) the results. Graphing, printing, and file management capabilities are included

as features, and there is a section-specific Help file. Shading calculations and economic evaluations are also included. The economic evaluations take into account the time-of-day energy production and demand values of BIPV.

Three primary factors influence the performance of BIPV: location, building design, and PV system specifications. These three factors are covered in three different sections: Location, Building, and PV System. In these sections, the user interactively develops a model of the BIPV project. In the Results section, the user can further investigate shading from individual surfaces and can evaluate the results in terms of economics.

The software includes typical meteorological year weather data (TMY2) for 239 U.S. cities, compressed into representative monthly average values and eliminating nighttime hours, when there is no incoming solar radiation to the PV panels. We also included weather data for 450 international cities. A database of PV modules is included, with the ability to customize inputs. The amount of

energy produced on each surface of a building is determined using well-accepted engineering formulas from the University of Wisconsin-Madison College of Engineering's Solar Energy Lab. Calculations are performed on an hour-by-hour basis, with temperature and system efficiency corrections.

CUSTOMIZING CHOICES

The program allows for custom inputs of building data files, along with custom weather, economics, and PV system inputs. Complete project files, as well as individual section files, can be saved for future retrieval. Results can be viewed, saved, and printed out in tabular or graphical formats. Results include averaged monthly hour-by-hour energy analysis of the BIPV system, using averaged weather data, which can be viewed in either tabular or graphical format, selected by month or by surface. Weather data information can be viewed either as ambient temperature or as global horizontal radiation.

We designed the program so that users can input data and get results in various formats. Three-dimensional (3-D) visualizations can be created using Virtual Reality Markup Language (VRML), a simple text-based surface- and light-generation language. Sketch UP by Google performs a similar function. Energy Ideas has demonstrated attaching light levels from weather data files (both beam and diffuse) to the design mock-up, and visualizing the results with full 3-D walk-through possibilities.

To date, over 1,400 users have visited the download page. To further develop BIPV Designer, Energy Ideas is in the process of putting the software into the public domain and making it an Open Source Project. This will allow expert users to collaborate on the further development of BIPV Designer and add additional features to it. For example, pollution prevention calculators could be developed that can help users understand the environmental benefits of a proposed BIPV project. Specialized front ends, or customized screens, could be

adapted to use surface files from CAD programs, while results calculations can be used to produce customized sales quotes.

We at Energy Ideas are continually revisiting the features of BIPV Designer to try to make it as useful as possible to a wide audience. We hope to make it easier for architects, designers, and homeowners alike to include BIPV in their designing process.



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For more information:

- To download BIPV Designer, visit www.energyi.mccabe.net.
- To learn more about this Open Source project, contact Joseph McCabe at energyi@mccabe.net.